

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: WAN Kiu Sum

Student ID: 25025236d

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ~~192.168.13.1~~ S0/0/0 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ~~781~~ 29 0

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No, S0/0/1 was passive on R2) 2

What does this information tell you?

The passive interface does not send OSPF hello packets; therefore, an adjacency will never be formed.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

enable
configure terminal
router ospf 1
no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

R3 S0/0/1 → R2 S0/0/1 (serial link): Cost = 781, then, R2 S0/0/0 → 192.168.2.0/24 (GigabitEthernet): Cost = 1, So the total cost is 781 + 1 = 782

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____ OSPF default bandwidth is 100 Mbps. If a network has higher speeds (i.e., 1 Gbps) then the cost of that link will be 1, which provides no differentiation of the link costs.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ Cost of 192.168.3.0/24: R1 S0/0/1 (781) + R3 G0/0 (1) = 782.

_____ Cost of 192.168.23.0/30: R1 S0/0/1 (781) + R3 S0/0/1 (64) = 845.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562 › The cost for each serial link is 781. Since the 192.168.23.0/24 network uses two serial links, we can multiply this by 2. That would be $781 \times 2 = 1562$.

Step 3: Change the route cost.

- c. Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2

OSPF will always choose the least expensive route to travel to its destination. Using R2 as an example; the least expensive route would be the path from R1 via S0/0/0 to R2 then via S0/0/1 to R3, G0/0. The total cost for that path would be $781 + 781 + 1 = 1563$. The most direct path, R1-S0/0/1-R3-G0/0 has a cost of $1565 + 1 = 1566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID is used in determining the Designated Router (DR), and Backup Designated Router (BDR). If you are using Router ID's that are dynamically generated based on the status of one or more interfaces then this could cause problems with stability when those interfaces go down. If you use a loopback interface for generating your Router ID then your Router ID will be able to remain stable regardless of whether any of your other interfaces fail.

2. Why is the DR/BDR election process not a concern in this lab?

A DR/BDR election will only happen on a multi-access network (such as Ethernet) in this lab, we have all of our interfaces connected by point-to-point serial links; therefore, DR/BDR election is not required.

3. Why would you want to set an OSPF interface to passive?

If you set an interface to passive, it stops OSPF from sending hello messages through the interface. The goal of setting an interface to passive is to limit OSPF traffic across your LAN when there are no OSPF neighbors present, ultimately conserving bandwidth and enhancing security.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF areas. OSPF divides the network into several areas. The backbone area is area 0. OSPF's use of areas for network organization improves the scalability of the network by reducing the amount of data that needs to be routed through each device and the effect on the network when topology changes are made.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Router(config-router)# network 10.0.0.0 0.255.255.255 area 0

This command is used in Router Configuration Mode (NOT Interface Configuration Mode), and it will assign any interfaces that have an IP address in the 10.0.0.0/8 range to area 0 using a wildcard mask (0.255.255.255).

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP will be used, Routers select routes based on Administrative Distance (AD), not metric. The default AD values are: EIGRP(internal) = 90 (Lowest = most trusted), OSPF = 110, RIP = 120 (highest = least trusted). Since EIGRP has the lowest AD, its route will be installed in the routing table, regardless of the metric values.